

# **From Free Rider to Free Raider: How Non-Enforcement Cascades Destroy Anonymous Cooperation**

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## **Abstract**

The free rider exploits cooperation passively, consuming without contributing. This paper identifies a more corrosive agent: the free raider, who actively preys on cooperators precisely because their cooperative behavior creates exploitable vulnerability. I develop this distinction within the extended phenotype framework (Dawkins 1982; Lerer 2025a), arguing that environments of sustained non-enforcement transform free riding into free raiding through a predictable cascade. When common knowledge of non-enforcement is established (Pinker 2007), anonymous cooperation collapses not because cooperators stop wanting to cooperate, but because the expected cost of cooperation rises above the expected benefit. The driver offering a ride to a stranger, the landlord renting to an unknown tenant, the shopkeeper extending credit to a neighbor: all face predation risk that scales with the depth of non-enforcement common knowledge.

The paper situates the free raider within a broader taxonomy of exploitation strategies, drawing on Goodman's (2025) evolutionary account of "invisible rivalry" and the game-theoretic formalization of reciprocity capture (Lerer 2026b). Three levels of predation on cooperation emerge: passive parasitism (the classical free rider), relational exploitation disguised as cooperation (the invisible rival who builds trust networks to extract transgressive favors), and direct predation on anonymous cooperators (the free raider who targets strangers whose cooperative signals reveal exploitable vulnerability). The three share a common causal variable: the cost of exploiting, determined by the structure of formal and informal enforcement. When that cost falls, the population redistributes across the three strategies in predictable sequence.

I formalize the free raider cascade as a three-stage process: tolerance (the state omits enforcement of property or personal security crimes), predation shift (criminals target cooperators rather than non-cooperators because cooperative behavior signals exploitable trust and resources), and cooperation collapse (anonymous cooperation withdraws to zero, leaving only intermediated cooperation channeled through state programs, unions, criminal networks, or clientelistic brokers). Evidence from Argentina (2020-2025), Chile (2022-2026), and Spain (2023-2026) documents this cascade across housing, transport, and retail domains. The framework generates a counterintuitive prediction: restoring enforcement alone is insufficient

to rebuild anonymous cooperation once the cascade completes, because common knowledge of non-enforcement persists even after enforcement resumes. Rebuilding requires costly public signals that create new common knowledge of safety, the mirror image of the televised non-enforcement announcements that triggered the cascade.

*Keywords: Free raider, anonymous cooperation, common knowledge, non-enforcement cascade, extended phenotype, invisible rivalry, reciprocity capture, property rights, clientelism*

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## I. Introduction

Decades ago, in Buenos Aires, Argentina, offering a stranger a ride used to be unremarkable. Micro-acts of anonymous cooperation, invisible to economists and unrecorded in any dataset, constitute what Ostrom (1990) called the "quiet infrastructure" of social trust. They require no contract, no intermediary, no state enforcement. They require only a background expectation that the stranger will not assault you. Nowadays that is infrequent, but still exists.

That expectation has eroded. Between 2020 and 2025, Argentina experienced a 420% increase in property usurpations alongside a collapse in prosecution rates to near zero (Lerer 2026a). But the consequences extend far beyond housing. When everyone knows that everyone knows the state will not prosecute property crimes, the inference generalizes: if the state will not protect my apartment, will it protect me in my car? If courts take three years to evict an occupant, will police respond when a passenger assaults a driver? The common knowledge of non-enforcement, once established in one domain, diffuses memetically across all domains where anonymous cooperation depends on background enforcement credibility.

Olson (1965) formalized the free rider as the central obstacle to collective action: someone who benefits from public goods without contributing. The free rider is passive. A more corrosive agent operates in non-enforcement environments: the free raider, who actively preys on cooperators because their cooperative behavior creates exploitable vulnerability. The driver offering a ride exposes herself to confined space with a stranger. The landlord renting an apartment provides a stranger with keys to her property. The shopkeeper extending credit reveals cash flow patterns. Each act of anonymous cooperation generates information and access that a predator can exploit. In enforcement environments, the expected predation cost is low (police respond, courts adjudicate, criminals face sanctions). In non-enforcement environments, the expected predation cost rises without bound, and anonymous cooperation collapses.

This paper formalizes the free raider concept within the extended phenotype framework (Dawkins 1982; Lerer 2025a), locating it at the intersection of three literatures: Pinker's common knowledge theory (2007), Ostrom's institutional analysis of cooperation (1990), and evolutionary psychology of invisible rivalry (Goodman 2025). I argue that sustained non-enforcement triggers a predictable cascade from free riding to free raiding to cooperation collapse, and that this cascade explains why clientelistic societies exhibit systematically lower levels of anonymous cooperation than enforcement societies, even controlling for GDP per capita and cultural variables. The framework connects to a companion analysis of reciprocity capture in organizational settings (Lerer 2026b), revealing that the three levels of exploitation, passive parasitism, relational predation, and direct predation on strangers, share a common causal structure despite operating at different scales.

## **II. From Free Rider to Free Raider: A Typology of Exploitation**

### ***II.A. The Parasitic Niche: Classical Free Riding***

The free rider occupies a parasitic niche: he benefits from the cooperative system without damaging it sufficiently to destroy it. Dawkins (1979) noted that successful parasites do not kill their hosts; they extract resources while keeping the host functional. The free rider on public transit, the neighbor who never contributes to building maintenance, the colleague who coasts on team effort: all exploit cooperation without eliminating it. The system tolerates them because their marginal cost is distributed across many cooperators.

The classical literature on collective action (Olson 1965; Hardin 1968) treated free riding as the fundamental threat to cooperation. Ostrom's (1990) empirical program demonstrated that communities develop graduated sanctions to manage free riders without destroying the cooperative system. The key insight is that the parasite needs a living host. The free rider who destroys the cooperative system eliminates her own food source. This self-limiting dynamic explains why societies tolerate moderate levels of free riding indefinitely: it is a chronic condition, not a terminal one.

### ***II.B. The Predatory Niche: Free Raiding***

The free raider occupies a predatory niche. Rather than passively consuming the fruits of cooperation, the free raider actively targets cooperators because their cooperative behavior signals vulnerability. Consider three cases.

**The benevolent driver.** A woman offers a ride to a stranger at a bus stop. The stranger, once inside the vehicle, assaults her and steals the car. The cooperative act (offering the ride) created the vulnerability (confined space, possession of a valuable asset). A non-cooperator who drove past the stranger faced zero predation risk. The free raider selectively targets the cooperator, not the non-cooperator.

**The trusting landlord.** A property owner rents an apartment to a young couple who present well, pay the first two months, then stop paying and refuse to leave. Under non-enforcement regimes (Argentina's near-zero eviction rate, Spain's three-property threshold), the landlord faces years of litigation while bearing mortgage, tax, and maintenance costs. The cooperative act (renting to strangers via market transaction) created the vulnerability. Landlords who kept properties empty or sold them faced zero okupacion risk.

**The generous shopkeeper.** A corner store owner in a Buenos Aires barrio extends informal credit to regular customers. One customer accumulates debt, then sends associates to intimidate the shopkeeper into canceling it. The cooperative act (extending credit based on personal trust) created the vulnerability. Large chains that accept only cash or card, with security cameras and corporate legal departments, face negligible risk from this predation.

In each case, the predator exploits a specific feature of anonymous cooperation: the absence of institutional intermediation. Anonymous cooperation operates without contracts, guarantees, or enforcement mechanisms; it relies on social trust and background state enforcement. When background enforcement collapses, anonymous cooperation becomes the optimal predation target because it offers value (the car, the apartment, the credit) with minimal defensive infrastructure.

### ***II.C. The Exploitation Spectrum: Three Levels of Predation on Cooperation***

The free rider and the free raider represent two extremes of a spectrum. Between them lies a third strategy that the compliance and organizational behavior literature has begun to formalize: the invisible rival (Goodman 2025), whose organizational expression I have analyzed elsewhere as reciprocity capture (Lerer 2026b). Mapping the three strategies reveals a unified logic of exploitation that operates across scales, from interpersonal interaction to institutional design.

Level 1: passive parasitism. The classical free rider consumes without contributing. The mechanism is non-contribution: the parasite sits on the cooperative surplus without reciprocating. The target is the public good itself (non-excludable by definition), not any specific cooperator. The effect on the host system is degradation without destruction: the cooperative system functions below optimum but remains viable. Policy response is well understood since Ostrom (1990): monitoring, graduated sanctions, reputation mechanisms. The cost of exploitation is low (merely abstaining from contribution), the return is moderate (share of public good), and the risk is manageable (social disapproval, mild sanctions).

Level 2: relational predation disguised as cooperation. Goodman (2025) argues that when human ancestors eliminated overt alpha dominance hierarchies, they replaced visible selfishness with something subtler: the capacity to move through society while planning and coordinating covertly. Humans cooperate until they identify an opportunity to exploit; the sophistication lies in maintaining the appearance of cooperation throughout. This is the "invisible rival," and Goodman's core claim is that it represents not a pathology but a universal human capacity, suppressed by enforcement and reputation costs, latent in every cooperator.

The organizational expression of invisible rivalry is reciprocity capture (Lerer 2026b). An auditor develops close working relationships with the managers she audits over eight years. Favors accumulate: career support, project assignments, political protection during organizational conflicts. When a manager requests "flexibility" on a finding, the auditor faces a social cost of refusal (violation of reciprocity obligation, loss of network standing, reputational damage as "disloyal") that may exceed the formal sanction for compliance failure. The auditor cooperates selectively: she cooperates with the manager (transgressing the formal rule) while presenting herself as cooperating with the organization (fulfilling her audit function). Both parties benefit; the cost falls on an absent third party (the organization, its shareholders, the public). Sai et al. (2026) demonstrate that this pattern is not learned in adulthood but consolidated by age five: children who receive favors lie to benefit their benefactors even when lying imposes personal costs (48% vs. 20%, OR = 3.66), harms innocent third parties (44% vs. 10%, OR = 7.04), or occurs without the benefactor present to verify compliance (37% vs. 10%).

The invisible rival's target is not the public good (as with the free rider) but the cooperator's trust within a relationship. The mechanism is not non-contribution but active exploitation of accumulated social capital. And critically, the invisible rival needs the cooperative relationship to remain functional: she continues cooperating in most dimensions (competent audit work, reliable professional performance) while extracting rents in specific high-value moments. The host survives; the extraction is selective and sustainable.

Level 3: direct predation on anonymous cooperators. The free raider targets strangers whose cooperative behavior reveals exploitable vulnerability. Unlike the invisible rival, who invests in building a relationship before exploiting it, the free raider exploits the generic signals of cooperation: trust, openness, asset exposure. No prior relationship is required. No sustained interaction is anticipated. The predation is one-shot, extractive, and frequently destructive. The benevolent driver, the trusting landlord, and the

generous shopkeeper are all targeted not because of who they are but because of what their cooperative behavior reveals: available resources and insufficient defenses.

Goodman's (2025) evolutionary framework illuminates why cooperation signals function as vulnerability markers. In ancestral environments, cooperation required observable actions: sharing food, tending the sick, participating in hunts. These signals were honest (costly to fake) and served as reliable indicators of prosocial disposition. Goodman identifies a critical asymmetry: the same signals that enabled cooperation also revealed who could be exploited, because cooperators by definition have resources (they share), are willing to engage with others (they approach), and prioritize social harmony over confrontation (they cooperate rather than fight). In small-scale societies, reputation costs constrained exploitation: everyone knew everyone, and predators faced social exclusion. In anonymous urban environments, reputation costs collapse. The free raider operates where no one knows her name, targeting cooperators whose signals were designed by evolution for small groups with memory, not for cities with anonymity.

**Table 1. Three Levels of Predation on Cooperation**

| Dimension                   | Level 1: Free Rider                       | Level 2: Invisible Rival                                            | Level 3: Free Raider                                                            |
|-----------------------------|-------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Strategy                    | Passive parasitism                        | Relational exploitation disguised as cooperation                    | Active predation on anonymous cooperators                                       |
| Target                      | Public goods (non-excludable)             | Cooperator's trust within a relationship                            | Cooperators identifiable by cooperative signals                                 |
| Mechanism                   | Non-contribution                          | Accumulation and exploitation of reciprocity obligations            | Exploitation of trust, access, and information revealed by cooperative behavior |
| Prior relationship required | No                                        | Yes (investment in building social capital)                         | No (targets strangers)                                                          |
| Effect on host              | Parasitic: degrades but sustains          | Selective extraction: host survives, rents captured                 | Predatory: destroys the cooperation it exploits                                 |
| Cooperator response         | Toleration (marginal cost distributed)    | Complicity (reciprocity obligation experienced as moral duty)       | Withdrawal (cooperation becomes irrational)                                     |
| Systemic outcome            | Suboptimal cooperation persists           | Institutional integrity erodes from within                          | Anonymous cooperation collapses; intermediated cooperation replaces it          |
| Enforcement threshold       | Low $k$ sufficient to contain             | Moderate $k$ plus reputation costs required                         | High $k$ required; common knowledge of enforcement essential                    |
| Key literature              | Olson (1965), Ostrom (1990)               | Goodman (2025), Lerer (2026b)                                       | This paper; Lerer (2026a)                                                       |
| Example                     | Tenant who underpays building maintenance | Auditor who overlooks findings for manager who supported her career | Okupa who occupies property; passenger who assaults benevolent driver           |

Source: Author's elaboration based on Olson (1965), Ostrom (1990), Dawkins (1982), Goodman (2025), Lerer (2026a, 2026b).

#### **II.D. The Common Variable: Cost of Exploitation**

The three strategies share a common causal variable: the cost of exploiting ( $k$ ), determined by the structure of formal and informal enforcement. Under high enforcement ( $k$  high), all three strategies are constrained. Free riding persists because it is difficult to detect and mildly costly to sanction. Invisible rivalry is suppressed because reputation costs remain operative (Goodman 2025) and formal sanctions deter the most egregious reciprocity-driven transgressions (Lerer 2026b). Free raiding is effectively eliminated because police respond, courts adjudicate, and predators face imprisonment.

When enforcement declines, the population redistributes across the three strategies in a predictable sequence. First, free riding increases at the extensive margin: individuals who contributed reluctantly under

enforcement stop contributing. Second, invisible rivalry intensifies within organizations as reciprocity costs dominate formal sanctions (the mechanism formalized in Lerer 2026b, where transgression becomes rational when the social cost of violating reciprocity,  $\beta \times \Delta$ , exceeds the expected formal sanction  $p \times S$ ). Third, and only under sustained, publicly known non-enforcement, free raiding emerges as a viable strategy because predation on anonymous cooperators carries near-zero expected cost.

The sequence matters. Free raiding does not appear immediately when enforcement weakens; it requires the prior establishment of common knowledge that enforcement has collapsed (Pinker 2007). This is why isolated enforcement failures (a corrupt judge, an overwhelmed police precinct) do not trigger the free raider cascade. Only systematic, publicly communicated non-enforcement creates the conditions for the third level. The televised eviction moratorium, the unenforced judicial order, the publicized okupacion statistics: these are the common knowledge signals that activate the transition from Level 2 (invisible rivalry within institutions) to Level 3 (open predation on anonymous cooperators).

The implication for the literature on cooperation is pointed. Olson (1965), Hardin (1968), and their successors diagnosed the free rider as the central obstacle to collective action. This diagnosis is incomplete. The free rider is manageable; societies have evolved mechanisms (reputation, graduated sanctions, reciprocity norms) to contain parasitism below system-destroying levels. The lethal threat is the free raider, and the literature's obsession with free riding has obscured the more dangerous dynamic: the cascade from parasitism through relational exploitation to open predation, driven by the common variable of enforcement cost. Goodman's evolutionary psychology, my formalization of reciprocity capture, and the free raider cascade documented here are three expressions of the same underlying structure, operating at the scales of dyadic interaction, organizational networks, and anonymous urban society respectively.

### **III. The Non-Enforcement Cascade**

The transition from free riding to free raiding follows a three-stage cascade. The cascade is not inevitable; it requires sustained non-enforcement accompanied by common knowledge creation. Isolated enforcement failures do not trigger the cascade. Only systematic, publicly known non-enforcement creates the conditions for predation shift.

#### ***III.A. Stage 1: Tolerance***

The cascade begins when the state publicly signals non-enforcement. Pinker's (2007) common knowledge framework explains why public signals differ qualitatively from private enforcement failures. When Argentina's President Fernandez decreed COVID-19 eviction moratoria via nationally televised address in March 2020, he did not merely change policy; he created common knowledge that the state would not enforce property rights. Potential predators learned the policy, learned that potential victims learned it, and learned that police and judges learned it. This infinite mutual knowledge transformed the enforcement landscape.

Tolerance need not be explicit. Spain's Ley de Vivienda (May 2023) did not announce "we will not enforce property rights." It introduced rent controls, penalties for "empty" homes, and eviction restrictions for landlords with three or more properties. The practical effect, however, was identical: potential okupas could infer from public law that landlords with multiple properties could not evict them. The information asymmetry shifted: okupas knew which landlords were defenseless (public property registries reveal ownership patterns), while landlords could not identify which potential tenants were okupas.

Chile's Boric combined explicit and implicit tolerance. The explicit channel: "no evictions without housing solutions" (televised campaign debates, 2021). The implicit channel: eight judicial eviction orders affecting 14,000+ people remained unexecuted as of September 2025, creating common knowledge that even court orders would not be enforced. When the state's own judiciary issues orders that the state's own executive ignores, the common knowledge of non-enforcement becomes maximally credible.

### ***III.B. Stage 2: Predation Shift***

Once common knowledge of non-enforcement is established, rational predators shift targeting from non-cooperators to cooperators. This is the critical transition. In enforcement environments, predators face roughly equal expected costs regardless of target: police respond, courts adjudicate, prison awaits. Target selection is driven by opportunity and reward. In non-enforcement environments, predators face differential expected costs: targeting cooperators yields higher rewards with lower resistance.

The mechanism operates through what I call the cooperator's signal. Anonymous cooperation emits signals that predators can decode: the driver who stops for a stranger signals trust, vehicle possession, and willingness to engage with unknowns. The landlord who lists an apartment on the rental market signals property ownership, regular income, and willingness to transact with strangers. The shopkeeper who extends credit signals cash availability, community embeddedness (making flight unlikely), and preference for social harmony over confrontation.

Goodman's (2025) evolutionary analysis explains why these signals are so reliably exploitable. Cooperation in ancestral environments required honest, costly signaling: sharing food meant having food, approaching strangers meant having social confidence, extending trust meant having resources worth risking. These signals evolved under conditions where exploitation was constrained by group memory and reputation. The free raider exploits a mismatch between the signaling environment in which cooperation evolved (small groups, universal surveillance, long memory) and the environment in which it now operates (large cities, anonymity, institutional dependence). The cooperator's signal is an honest signal in the wrong environment, like a tropical bird's plumage displayed before a predator that never existed in its ancestral habitat.

Goodman (2025) makes a further point directly relevant to the predation shift. In his analysis of "The Problem of Opportunity," he argues that cooperation and cooperators must be analytically distinguished. Many individuals cooperate only because they lack the opportunity to exploit; punishment theories, he notes, "must be silent about internal states." The predation shift is, in this framework, a mass revelation of latent exploitation preferences. Individuals who cooperated under enforcement, not from disposition but from constraint, discover that non-enforcement has opened an opportunity window. The cascade does not require an invasion of external predators; it converts existing population members from cooperators to raiders at the extensive margin. The free raider was always present, latent, waiting for the cost of exploitation to fall below the threshold of activation.

Predators optimize against cooperation signals. In Buenos Aires, organized crime networks (known as "entraderas") specifically target homeowners returning from supermarkets; the shopping bags signal both resources and distraction. In Barcelona, okupacion networks consult public property registries to identify landlords with three or more properties, knowing these landlords cannot legally evict them. In Chile, Los Trinitarios established territorial control over campamentos, charging "rent" to occupants; the cooperative act of settling in an informal community created vulnerability to organized extraction.

The predation shift inverts Maimonides' charity hierarchy at a deeper level than Lerer (2026a) identified. Not only are anonymous market transactions stigmatized as "speculation"; anonymous charitable acts (the ride, the informal credit, the neighborly favor) become actively dangerous. Maimonides ranked anonymous giving highest because it preserved the dignity of both giver and receiver. The free raider transforms anonymous giving from the highest form of charity into the highest form of vulnerability.

### ***III.C. Stage 3: Cooperation Collapse and the Conversion Mechanism***

When predation risk exceeds cooperation benefit, anonymous cooperation withdraws. The process follows a tipping point dynamic analogous to Schelling's (1978) neighborhood segregation model. Below the tipping point, most people still cooperate and predation remains rare; a few bad experiences are absorbed as "bad luck." Above the tipping point, predation becomes common enough that cooperators revise their priors: offering rides becomes irrational, renting to strangers becomes irrational, extending credit becomes irrational. Cooperation collapses rapidly once the tipping point is crossed.

A critical acceleration mechanism operates at this stage: the conversion of free riders into free raiders. Under enforcement, the free rider's optimal strategy is parasitism, consuming the cooperative surplus without actively destroying it. Under non-enforcement, the same individual discovers that predation yields higher returns at zero additional cost. The neighbor who never contributed to building maintenance (Level 1) graduates to occupying a vacant apartment in the building (Level 3). The colleague who coasted on team effort graduates to embezzling team resources. The conversion does not require a change in preferences, only a change in opportunity cost. Goodman (2025) would predict exactly this: the "invisible rival" lurking within every cooperator requires only a reduction in exploitation cost to activate.

The conversion mechanism explains why cooperation collapse, once initiated, accelerates convexly. Each new raider reduces the cooperator population (fewer targets) while increasing the raider population (more predators per target). But the raiders are not invading from outside; they are converting from within the population. The free rider who becomes a free raider was previously stabilizing the system (parasites need living hosts); now she destabilizes it (predators destroy hosts). The system loses its internal brake.

The collapse is selective. Anonymous cooperation disappears; intermediated cooperation survives or expands. When landlords stop renting to strangers, they rent through agencies with guarantee services (intermediated by a commercial entity). When drivers stop offering rides, ride-sharing platforms with identity verification and GPS tracking expand (intermediated by technology). When shopkeepers stop extending credit, formal consumer finance companies fill the gap (intermediated by financial institutions). Each transition replaces anonymous, costless cooperation with intermediated, costly cooperation. The intermediary captures rents; the cost of cooperation rises; the poorest users, who cannot afford intermediation fees, lose access entirely.

This is where the free raider cascade connects to clientelism. As anonymous cooperation collapses, demand for intermediated cooperation rises. The state, unions, punteros, and criminal networks all offer intermediation services: the state provides housing (via visible programs with electoral payoff), unions negotiate collective agreements (capturing dues), punteros distribute benefits (capturing votes), and criminal networks provide "protection" (capturing tribute). The collapse of anonymous cooperation is the precondition for clientelistic capture. Clientelism does not merely exploit a pre-existing dependency; it creates the conditions for dependency by tolerating the free raider cascade that destroys the anonymous cooperation which would otherwise make clientelistic intermediation unnecessary.



## IV. Formalization

### IV.A. Basic Model

Let  $c$  represent the cost of cooperating (time, effort, exposure risk) and  $b$  the benefit (social reciprocity, moral satisfaction, practical return). In enforcement environments, the expected predation loss is  $p_e * L$ , where  $p_e$  is the probability of predation given enforcement and  $L$  is the loss if predation occurs. Anonymous cooperation is rational when:

$$b > c + p_e * L$$

In non-enforcement environments, the probability of predation rises to  $p_n \gg p_e$  because: (a) existing predators face lower costs and increase frequency, (b) marginal individuals who would not predate under enforcement shift to predation (the extensive margin, i.e., the FR-to-FRA conversion), and (c) common knowledge of non-enforcement coordinates predation by reducing uncertainty about state response. Cooperation is rational when:

$$b > c + p_n * L$$

The cascade triggers when  $p_n * L$  exceeds  $(b - c)$ , the net cooperation surplus. Two features accelerate the cascade beyond what a simple parameter shift would predict. First,  $p_n$  is endogenous: as cooperators withdraw, the remaining cooperators face higher predation concentration (fewer targets, same number of predators), further increasing  $p_n$ . This is a positive feedback loop. Second,  $L$  is endogenous to non-enforcement: when courts do not adjudicate, losses cannot be recovered; when police do not respond, physical harm goes uncompensated. Both  $p_n$  and  $L$  increase simultaneously under non-enforcement, creating a convex acceleration that explains why cooperation collapse, once initiated, completes rapidly.

### IV.B. Three-Species Dynamics

The three-level taxonomy suggests a population structure with cooperators (C), free riders (FR), and free raiders (FRA). The pairwise interaction payoffs illuminate why the free rider, paradoxically, has an objective interest in maintaining the cooperative ecosystem.

When two cooperators interact, both obtain the cooperation surplus  $(b - c)$ . When a cooperator meets a free rider, the cooperator pays the cooperation cost while the rider extracts benefit without reciprocating; the system degrades but survives. When a cooperator meets a free raider, the cooperator loses not merely the cooperation cost but exposed assets ( $L \gg c$ ); the interaction is destructive rather than parasitic. When two free riders meet, no interaction occurs (there is no cooperation to extract). When a free rider encounters a free raider, the rider is a suboptimal target: she does not emit cooperation signals, does not expose assets, and offers lower expected return than a cooperator. When two free raiders meet, the interaction is costly for both (intraspecific competition over shrinking prey).

The critical insight is that the free rider needs a functioning cooperative ecosystem to survive. Parasites require healthy hosts. When the free raider drives the cooperator population below a viability threshold, the free rider loses her food source before the free raider does, because riders need sustained cooperation while raiders can operate on the last remaining cooperators. The free rider thus has an objective (if unintentional) interest in constraining the free raider, a dynamic absent from the classical Olson framework that treats all exploitation as equivalent.

Under enforcement ( $k$  high), FRA is a dominated strategy: the expected cost of predation exceeds the return. The game reduces to C vs. FR, Olson's classical dilemma, manageable through monitoring and graduated sanctions. Under non-enforcement ( $k$  approaching zero), FRA dominates FR: the raider can do everything the rider does plus predate, at no additional cost. And FRA dominates C trivially when  $L - k > b - c$ , a condition satisfied by any nontrivial predation yield. The equilibrium under sustained non-enforcement is therefore extinction of C, followed by extinction of FR, leaving only FRA, which then faces the ironic problem of having destroyed its own prey base.

#### ***IV.C. Hysteresis***

The model predicts a hysteresis effect. Restoring enforcement (reducing  $p$  back toward  $p_e$ ) does not immediately restore cooperation, because cooperators have updated their priors based on experienced predation. Even after enforcement resumes, the subjective probability of predation  $p_{\text{subjective}}$  remains elevated relative to  $p_e$  for a period proportional to the duration and severity of the non-enforcement episode. This hysteresis explains why countries that restore property rights enforcement (e.g., post-crisis Argentina under Milei, 2024-2025) observe slow recovery of rental supply and anonymous cooperation despite rapid enforcement policy changes.

The hysteresis is reinforced by the conversion mechanism. Former free riders who converted to free raiding during the non-enforcement period do not automatically revert when enforcement resumes. Predation may have generated material gains, social networks, and skill sets (how to identify vulnerable targets, how to navigate legal loopholes) that persist beyond the enforcement change. Reconversion requires not merely restoring the cost of exploitation ( $k$ ) but exceeding the sunk investments in raiding infrastructure, a higher threshold than the original activation cost.

### **V. Preliminary Evidence**

Systematic measurement of anonymous cooperation is inherently difficult; the phenomenon is defined by its informality and invisibility. I rely on proxy indicators across three domains.

#### ***V.A. Housing: Rental Supply Contraction***

Argentina's formal rental listings in Buenos Aires declined approximately 35-40% between 2020 and 2023 (multiple real estate portal estimates). Spain's rental supply contracted 12% in Barcelona and 8% in Madrid between 2023 and 2025 following the Ley de Vivienda (Idealista, 2025). In both cases, landlords withdrew from anonymous market transactions (renting to strangers) in favor of non-rental uses (sale, tourist rental, family use) or intermediated transactions (guarantee agencies, corporate rentals). The withdrawal is a direct manifestation of cooperation collapse: landlords who previously trusted anonymous market mechanisms no longer do so.

#### ***V.B. Transport: Ride-Sharing and Stranger Avoidance***

Argentine ride-sharing platform usage increased 140% between 2019 and 2024, while informal ride-sharing (offering rides to strangers, carpooling without platform intermediation) declined. Precise measurement of informal ride-sharing decline is unavailable, but survey evidence from CESBA (Centro de Estudios de Sociedad, Bienestar y Accion, Buenos Aires, 2024) reports that 67% of respondents "never" offer rides to strangers, up from 43% in 2018. The shift from anonymous to intermediated transport

cooperation is consistent with the free raider cascade: platform intermediation provides identity verification, GPS tracking, and institutional accountability that anonymous cooperation cannot.

### ***V.C. Retail: Credit and Trust Contraction***

Argentina's informal retail credit (*fiado*) declined in Buenos Aires barrios during the same period. CAME (Confederacion Argentina de la Mediana Empresa) reported in November 2024 that 72% of small retailers had reduced or eliminated informal credit, citing "insecurity and non-payment" as primary reasons. The replacement is formal consumer finance: Mercado Credito, personal loans, and credit card installment plans (*cuotas*). Each replacement channels cooperation through an intermediary that captures fees, excludes the poorest consumers (who lack credit histories), and eliminates the social bond that informal credit created.

## **VI. Implications**

### ***VI.A. For the Extended Phenotype Framework***

The free raider cascade extends the extended phenotype framework (Lerer 2025a, 2025b, 2026a) in three directions. First, it identifies the transmission mechanism between property rights collapse and generalized cooperation collapse. The Road to Connivance paper (Lerer 2026a) documented how common knowledge of non-enforcement enables coordinated property occupation. The free raider cascade shows that the damage extends beyond property: once common knowledge of non-enforcement is established in any domain, it diffuses to all domains where anonymous cooperation depends on background enforcement credibility. Property rights collapse is not merely a housing problem; it is a cooperation infrastructure problem.

Second, it explains why clientelistic societies exhibit lower anonymous cooperation than enforcement societies even at comparable income levels. The conventional explanation attributes low cooperation to "low social trust" or "weak civic culture," treating trust as an exogenous cultural variable. The free raider cascade offers a structural alternative: anonymous cooperation is low because the non-enforcement environment makes it rationally dangerous. Trust is not culturally absent; it is structurally suppressed. Restoring enforcement should, over time, restore anonymous cooperation, a prediction testable in Argentina's post-2024 enforcement reforms.

Third, it reveals the self-reinforcing nature of clientelistic lock-in. The connivance equilibrium (Lerer 2026a) persists not merely because politicians benefit from it electorally, but because it destroys the anonymous cooperation that would make clientelistic intermediation unnecessary. Citizens who cannot safely rent to strangers, offer rides to strangers, or extend credit to strangers become dependent on intermediated cooperation provided by the state, unions, and informal networks. The clientelistic equilibrium reproduces itself by eliminating its own alternatives.

### ***VI.B. For the Literature on Cooperation and Exploitation***

The three-level taxonomy reframes the relationship between cooperation research and its policy implications. The classical collective action literature (Olson 1965; Hardin 1968) focused on Level 1 (free riding) and generated Level 1 solutions: monitoring, sanctions, incentive alignment. The organizational behavior and compliance literature (Bazerman and Gino 2012; Trevino et al. 2006) has increasingly recognized Level 2 (reciprocity capture, normative conflict) and generated Level 2 solutions: rotation

policies, common knowledge mechanisms, externalized reporting channels. The present paper identifies Level 3 (free raiding on anonymous cooperators) and argues that it requires Level 3 solutions: restoration of background enforcement credibility through costly, visible, sustained public signals.

The policy error of applying Level 1 solutions to Level 3 problems is widespread. Governments that respond to okupacion with "better rental incentives" (Level 1: adjust the free rider's cost-benefit calculation) or "cultural campaigns for housing respect" (Level 1: normative suasion) miss the point: the problem is not that free riders exploit the housing system but that free raiders destroy the willingness to participate in it. Similarly, organizations that respond to reciprocity capture with "more training" or "stronger codes of ethics" (Level 1 solutions) fail because the problem operates at Level 2, where structural fragmentation and common knowledge mechanisms are required (Lerer 2026b). Matching the solution level to the problem level is a precondition for effective institutional design.

### ***VI.C. For Evolutionary Psychology***

Goodman's (2025) "invisible rivals" thesis gains empirical traction from the free raider cascade. If invisible rivalry is a universal human capacity (suppressed by enforcement and reputation, not absent from any population), then non-enforcement episodes function as natural experiments that reveal the latent distribution of exploitation preferences. Argentina 2020-2025 is not a story about a peculiarly corrupt population; it is a demonstration of what happens when enforcement collapses in any human society. The prediction is generalizable: any jurisdiction that creates common knowledge of non-enforcement will observe the same cascade, regardless of cultural context or GDP per capita. Spain and Chile provide partial confirmation across very different cultural settings.

The developmental evidence from Sai et al. (2026) reinforces this point. If reciprocity-driven dishonesty is consolidated by age five across cultures, the substrate for both invisible rivalry (Level 2) and free raiding (Level 3) is universal. What varies is not the human capacity for exploitation but the institutional structure that constrains it. Enforcement does not create moral behavior; it channels universal exploitation capacity toward less destructive strategies (free riding rather than free raiding, invisible rivalry rather than open predation). Removing enforcement does not create immoral behavior; it reveals the full range of exploitation strategies that enforcement had suppressed.

## **VII. Conclusion**

The free rider is a nuisance; the free raider is a systemic threat. The distinction matters because the policy responses differ qualitatively. Free riding can be managed through monitoring, incentives, and graduated sanctions (Ostrom 1990). Free raiding requires restoring the background enforcement credibility that makes anonymous cooperation viable. Half-measures fail: selective enforcement that is not publicly known does not rebuild common knowledge of safety. Only costly, visible, sustained enforcement creates the new common knowledge necessary to restart anonymous cooperation.

Between these extremes, the invisible rival (Goodman 2025) and reciprocity capture (Lerer 2026b) operate within organizations and trust networks, exploiting relationships rather than anonymity. The three levels form a unified exploitation spectrum sharing a common causal variable: the cost of exploiting, set by formal and informal enforcement. Lowering that cost redistributes the population across levels in predictable sequence; raising it compresses exploitation back toward Level 1, where it is chronic but manageable.

The cascade from tolerance to predation shift to cooperation collapse follows a predictable sequence, accelerated by the endogeneity of both predation probability and loss magnitude under non-enforcement, and by the conversion of free riders into free raiders at the extensive margin. The hysteresis effect implies that the cost of allowing the cascade to complete far exceeds the cost of preventing it: rebuilding anonymous cooperation after collapse requires sustained enforcement over years or decades, while preventing collapse requires merely maintaining existing enforcement.

Argentina's 2020-2025 trajectory offers a natural experiment. If the Milei administration's enforcement reforms (2024-2025) are sustained, the model predicts gradual recovery of rental supply, informal cooperation indicators, and social trust metrics over a 3-5 year horizon, with faster recovery in domains where enforcement is most visible (housing, transport) and slower recovery in domains where enforcement remains ambiguous (retail credit, neighborly cooperation). Failure to observe recovery within this timeframe would require revision of the hysteresis assumptions.

The road to connivance, I argued elsewhere, is paved with public announcements, inverted charity, and strategic omission. The road back requires public enforcement, restored anonymity, and rebuilt trust. The free raider cascade explains why the road back is longer than the road there: destroying cooperation is fast; rebuilding it is slow. Recognizing this asymmetry is the first step toward calibrating the institutional investment required.

## **Disclosure and Methodological Note**

I used large language models (Claude, Anthropic) as computational aids for structuring the game-theoretic formalization in Section IV and for literature synthesis. My training is in law and business, not in formal mathematics or evolutionary biology; AI assistance enabled me to express intuitions derived from three decades of legal practice in the notation conventions of evolutionary game theory. The theoretical framework, case analysis, policy implications, and all substantive interpretations are entirely my own. Any errors in formalization are mine.

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